

Data Collection and Preprocessing

STC224

Introduction to Data Collection

- **What is Data Collection?**
- The systematic process of gathering information to answer research questions or test hypotheses.
- Forms the foundation of all data analysis.
- Poor data collection = poor analysis outcome (“garbage in, garbage out”).

Purpose of Data Collection

1. To ensure data is accurate, relevant, and reliable.
2. To support decision-making and statistical inference.
3. To understand patterns and trends in a population or sample.
4. To ensure the analysis aligns with the research goals.

Types of Data

1. Qualitative Data

- Non-numerical (e.g., opinions, descriptions).

2. Quantitative Data

- Numerical values.
- **Discrete:** countable (e.g., number of students).
- **Continuous:** measurable (e.g., height, weight).

Sources of Data

- **Primary Data**

- Collected first-hand by the researcher.
- Examples: surveys, interviews, experiments.

- **Secondary Data**

- Obtained from already existing sources.
- Examples: government reports, databases, academic journals.

Data Collection Methods

- Surveys and questionnaires
- Interviews (structured/unstructured)
- Observation
- Experiments
- Administrative records
- Online data sources (APIs, websites, logs)

Challenges in Data Collection

- Non-response or incomplete data
- Sampling bias
- Measurement errors
- Recording and entry errors
- High costs and time required

Introduction to Data Preprocessing

- **What is Data Preprocessing?**

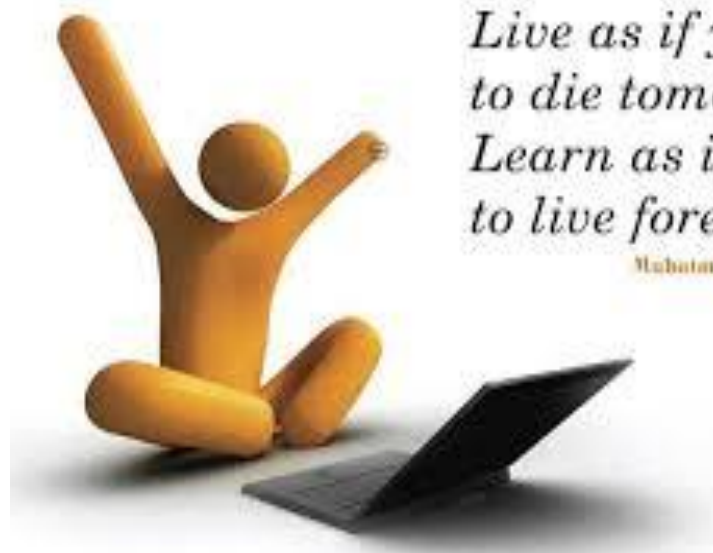
- A set of techniques used to prepare raw data for analysis.
- Ensures data is consistent, complete, and analysis-ready.
- Critical before using SPSS, STATA, or any statistical tools.

Stages of Data Preprocessing

- **Data inspection** (checking structure and quality)
- **Data cleaning** (fixing errors)
- **Data transformation** (standardization, normalization)
- **Data integration** (merging multiple datasets)
- **Data reduction** (simplification, feature selection)

Why Preprocessing Matters?

- Removes noise and inconsistencies
- Improves model accuracy
- Makes statistical analysis valid
- Helps software like SPSS read data correctly
- Ensures results reflect true patterns, not data errors



*Live as if you were
to die tomorrow.
Learn as if you were
to live forever.*

Mahatma Gandhi

What is Data Cleaning?

- Data cleaning is the process of detecting and correcting inaccurate or incomplete data.
Also known as data scrubbing.

Common Data Problems

- Missing values
- Duplicate records
- Outliers
- Inconsistent formats
- Typographical errors
- Incorrect coding of variables (e.g., Male = 1, Female = 2)

Handling Missing Data

- Methods
 - **Deletion:** remove missing cases
 - **Imputation:** replace with mean, median, mode
 - **Prediction models:** regression-based imputation
 - **SPSS tools:** *Replace Missing Values*

Handling Outliers

- Identify using:
 - Boxplots
 - Z-scores
 - Scatterplots
- Solutions:
 - Investigate for errors
 - Winsorization
 - Transformation
 - Exclusion (only if justified)

Data Validation

- Check **range** (e.g., SAT scores 200–800)
- Check **codes** (e.g., 1 = Home, 2 = Away)
- Check **consistency** across variables
- Confirm no **impossible values**

Data Transformation Techniques

- **Normalization:** scaling to a range (0–1)
- **Standardization:** converting to z-scores
- **Encoding:** converting categories to numbers
- **Log transformation:** reducing skewness

Data Integration

- Merging multiple datasets
- Dealing with differences in variable names
- Removing duplicated entries
- Ensuring consistent measurement units

Data Reduction Techniques

- Removing irrelevant variables
- Principal Component Analysis (PCA)
- Aggregation (summaries instead of raw)
- Binning numerical variables

Practical Examples in SPSS

- From the module outline:
- Enter data using **Variable View** and **Data View**
- Using SPSS tools for:
 - Detecting missing values
 - Transforming data
 - Viewing frequency tables
 - Checking ranges and outliers
 - Recoding variables

Summary

- Data collection is the foundation of data analysis.
- Preprocessing ensures data is clean and analysis-ready.
- Data cleaning addresses errors, missing values, and inconsistencies.
- SPSS supports these steps through built-in tools.
- Proper preparation increases accuracy and reliability of results.